

Task 3: Visualization Challenge

Power BI Sales Performance Dashboard

Organization: WeIntern Pvt Ltd

Tool: Microsoft Power BI Desktop

Source: Sample - Superstore

1. Objective

Task 3 presents the analyzed Sample Superstore data through a professional dashboard that communicates business performance using clear KPI cards and appropriate bar, line, pie, product, regional, and filtering visuals. The goal is to make the analytical results from Task 2 easier to understand and explore while maintaining clear labels, readable layout, and evidence-based interpretation.

2. Dataset and Analytical Context

Item	Value
Dataset	Sample - Superstore
Records	9,994 order-line records
Distinct orders	5,009
Customers	793
Products	1,850
Time period	2014-2017
Dashboard platform	Microsoft Power BI Desktop

The source data is stored at order-line level. Therefore, Total Orders is calculated using distinct Order ID rather than row count. Sales is treated as revenue. The dashboard visualizes the same verified business measures and aggregations established during the Task 2 analysis.

3. Data Preparation and Model Considerations

- Order Date and Ship Date were interpreted using U.S. month/day/year date logic.
- Order-based calculations use distinct Order ID to avoid counting multiple line items as separate orders.
- Revenue is based on the Sales field and units sold are based on Quantity.
- The dashboard uses the Sample Superstore table as its analytical source and applies Power BI measures and visual aggregations to it.

4. KPI Cards

KPI	Definition	Verified Result
Total Revenue	Sum of Sales	\$2,297,200.86
Total Orders	Distinct Order ID	5,009
Average Order Value	Total Revenue / Total Orders	\$458.61

Total Units Sold	Sum of Quantity	37,873
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The KPI cards provide an immediate summary of overall business scale. Revenue and units describe sales magnitude, while distinct orders and average order value provide transaction-level context. Viewing these measures together avoids interpreting performance from a single metric.

5. Dashboard Visualizations and Interpretations

5.1 Revenue by Category - Bar Chart

A bar chart is used because the visual compares a small set of discrete product categories. Technology generated \$836,154.03 in revenue and is the highest-revenue category. The comparison makes the relative contribution of the three categories easy to evaluate.

5.2 Monthly Sales Trend - Line Chart

A line chart is used to preserve chronological order and make changes over time visible. Monthly revenue varies substantially across the 2014-2017 period. November 2017 recorded the highest monthly revenue at \$118,447.83, while February 2014 recorded the lowest at \$4,519.89. The chart identifies peaks and declines but does not by itself establish the causes of those changes.

5.3 Revenue Share by Customer Segment - Pie Chart

A pie chart is appropriate here because the comparison contains only three customer segments. Consumer contributes the largest share of revenue at approximately \$1.16 million, followed by Corporate at approximately \$706.15 thousand and Home Office at approximately \$429.65 thousand. The limited number of segments keeps the proportional comparison readable.

5.4 Top 10 Products by Units Sold - Bar Chart

The Top 10 product chart ranks products by unit volume and limits the display to the strongest volume contributors to prevent an unreadable long-category chart. Staples is the highest-volume product at 215 units. Product volume should be interpreted separately from product revenue: the highest-revenue product is Canon imageCLASS 2200 Advanced Copier at \$61,599.82.

5.5 Revenue by Region

The regional visual compares revenue contribution across the four regions. The West leads with \$725,457.82, followed by the East at \$678,781.24, Central at \$501,239.89, and South at \$391,721.91. This view highlights geographic differences in revenue contribution without assuming that geography alone caused the differences.

6. Dashboard Layout and Interactivity

The dashboard is designed as a single-page business summary. KPI cards are positioned as headline metrics, followed by trend, category, product, customer-segment, and regional views. Year, Region, Category, and Segment slicers allow the user to narrow the dashboard to relevant subsets while retaining the same KPI and visual definitions.

Dashboard Element	Purpose
4 KPI cards	Headline performance summary

Revenue by Category	Category comparison
Monthly Sales Trend	Time-based trend
Revenue Share by Customer Segment	Simple proportional comparison
Top 10 Products by Units Sold	Product-volume ranking
Revenue by Region	Geographic comparison
Year slicer	Time filtering
Region slicer	Geographic filtering
Category slicer	Product-category filtering
Segment slicer	Customer-segment filtering

7. Key Findings

1. The dashboard summarizes \$2.30 million in revenue from 5,009 distinct orders and 37,873 units, with an average order value of \$458.61.
2. Technology is the strongest category by revenue at \$836,154.03.
3. November 2017 is the highest-revenue month in the dataset at \$118,447.83.
4. Consumer is the largest customer segment by revenue at approximately \$1.16 million.
5. The West is the highest-revenue region at \$725,457.82.
6. Staples leads unit volume at 215 units, while Canon imageCLASS 2200 Advanced Copier leads product revenue at \$61,599.82.

8. Recommendations

- Review revenue, orders, units, and average order value together when evaluating changes in dashboard performance.
- Track high-revenue products separately from high-volume products because the two rankings represent different commercial patterns.
- Use the region, category, segment, and year slicers to investigate whether headline patterns persist within narrower business groups.
- Use the dashboard as a summary layer and refer to the Task 2 analysis for deeper customer-level and trend interpretation.

9. Limitations and Assumptions

- The source dataset is at order-line level; distinct Order ID is used for order-based metrics.
- Sales is treated as revenue and Quantity is treated as units sold.
- The dashboard summarizes historical records in the supplied Sample Superstore dataset and is not a forecast.
- Visual differences describe associations in the supplied data and do not establish causal explanations.
- Profitability is not the primary focus of this Task 3 dashboard; revenue and sales-volume communication are emphasized.
- The PBIX is the authoritative interactive dashboard artifact; screenshots or report figures are static evidence of the completed work.

10. Task 3 Deliverables Coverage

Expected Task 3 Deliverable	Evidence in Submission
At least one bar chart	Revenue by Category; Top 10 Products by Units Sold
At least one line chart	Monthly Sales Trend
At least one pie chart	Revenue Share by Customer Segment
One dashboard mockup/layout	Single-page Power BI dashboard
Short interpretation for each visual	Section 5 of this report
KPI cards / business summary	Section 4 and Power BI dashboard
Clear chart titles and readable labels	Implemented in dashboard; documented in this report
Final report	This document / PDF export
Visual outputs / screenshots	To be included in repository Screenshots/ and Outputs/
Power BI dashboard file	Task3_Dashboard.pbix

11. Repository Deliverables

The final Task 3 repository should contain the Power BI dashboard file, the source dataset or a clear source reference, the final PDF report, genuine screenshots/export of the completed dashboard, visual outputs, and a Task 3 README with project summary and execution/opening instructions.

12. Conclusion

Task 3 converts the verified sales analysis into a concise visual decision-support layer. The dashboard satisfies the required visualization types while combining headline KPIs, time trends, category and product comparisons, customer-segment share, regional performance, and interactive filtering in one page. The report keeps interpretations evidence-based and separates observed patterns from recommendations.